ANSWER KEY & MARKING SCHEME SET A

Chapter 2: **Electricity: Circuits and Their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** battery	cells joined together
2	**(a)** closed circuit	complete path
3	**(c)** positive terminal	convention
4	**(c)** Plastic	does not conduct
5	**(d)** positive terminal is connected to the positive side of the cell	polarity matters
6	**(c)** direct current (DC)	flows one way
7	**(a)** Cells	low voltage
8	**(d)** contains dissolved substances	acids conduct
9	**(c)** circle with a cross or loop	symbol
10	**(b)** Cell, wires, bulb and switch	complete path

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A closed path along which electric current flows (2).	2
12	A cell is a single unit (1); a battery is two or more cells joined (1).	1+1
13	The circuit is broken so current cannot flow (2).	2
14	Conductors: copper, aluminium; insulators: rubber, plastic (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Connect cell terminal to bulb (1); bulb to switch (1); switch back to the other terminal (1).	1+1+1
16	Cell, switch, bulb in a loop (1.5); direction from positive terminal through the circuit to negative (1.5).	1½+1½
17	Make a circuit with a gap (1); place the material across the gap (1); bulb glows if conductor (1).	1+1+1
	OR: Conductors: copper, iron, aluminium (1.5); insulators: rubber, plastic, wood (1.5).	1½+1½
18	Rubber and wood are insulators (1.5); protect from shock (1.5).	1½+1½
	OR: Cell weak (1); fused filament (1); loose connection or open switch (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Conductors allow current; insulators do not (2). (ii) Copper, iron; rubber, plastic (1). (iii) Circuit with gap, test each material (2).	2+1+2
	OR: (i) Cell, bulb, switch in loop (1). (ii) Weak cell, fused filament, loose contact (2). (iii) Connect it to a fresh cell (1). (iv) Cell/battery (1).	1+2+1+1
20	(i) Symbols of components (1). (ii) Correct diagram (2). (iii) Longer line/bump (2).	1+2+2
	OR: (i) Insulators prevent shock (1). (ii) Metals (1). (iii) Water conducts so risk increases (1). (iv) Switch off mains, dry hands, use insulators (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 2: **Electricity: Circuits and Their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** positive (+) and negative (−)	electrodes
2	**(d)** open (OFF)	circuit is broken
3	**(d)** circuit diagram	standard symbols
4	**(d)** a conductor	a non-metal that conducts
5	**(d)** current heats the filament	heating effect
6	**(b)** closed	current flows
7	**(c)** fuse	wire melts on overload
8	**(a)** a poor conductor	salts make water conduct
9	**(d)** a gap	no connection
10	**(c)** ampere	A

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Thin wire in a bulb (1); it heats and glows (1).	1+1
12	Cell: long and short parallel lines (1); switch: break and lever or gap (1).	1+1
13	Plastic is an insulator and prevents shocks (2).	2
14	A representation of a circuit using standard symbols (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	DC flows in one direction, e.g. cells (1.5); AC changes direction, e.g. mains (1.5).	1½+1½
16	Do not touch bare wires (1); keep water away (1); use only cells in experiments (1).	1+1+1
17	Terminals reversed or not touching contacts (1.5); circuit incomplete (1.5).	1½+1½
	OR: Opens or closes the circuit (1.5); a light switch at home (1.5).	1½+1½
18	Correct: brighter (1.5); wrong: dim or off (1.5).	1½+1½
	OR: Salt dissolves into charged parts (1.5); distilled water has none (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Cell energy passes through filament (2). (ii) Cell, switch, bulb, reflector (2). (iii) Filament heats and glows (1).	2+2+1
	OR: (i) DC one direction; AC reverses (2). (ii) Cell; power plant (1). (iii) It is dangerous (2).	2+1+2
20	(i) Lemon and salt solutions conduct (1). (ii) Sugar does not produce charged particles (1). (iii) Pure water has none (1). (iv) To make it conduct (2).	1+1+1+2
	OR: (i) Heating and light effects (2). (ii) Heating/light (1). (iii) Heating (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.